

PRAVAL PRATAP SINGH

Ph.D. Research Scholar | Theoretical & Computational Chemist

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LinkedIn



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ORCID



Personal Website

RESEARCH EXPERIENCE

- Molecular Simulation & Membrane Biophysics:** Atomistic molecular dynamics of lipid bilayers and membrane proteins, with emphasis on lipid-water-solute coupling, sterol-mediated membrane organization, interfacial hydration dynamics, and membrane transport phenomena.
- Force-Field & Water-Model Benchmarking:** Systematic evaluation of lipid force fields and explicit water models for phosphatidylcholine membranes, focusing on structural, interfacial, thermodynamic, and dynamical properties.
- Computational Structural Biology & Molecular Recognition:** Structure-dynamics relationships in membrane transporters, particularly human SGLT2; lipid-protein coupling, internal hydration, protein-ligand recognition, and disease-associated protein conformational dynamics, including A β protofibrils and intrinsically disordered proteins.
- Quantum Chemistry, Functional Nanomaterials & Machine Learning:** DFT-based studies of adsorption, reaction mechanisms, electronic structure, and molecular sensing using heteroatom-doped fullerenes and borophene, integrated with cheminformatics and machine-learning approaches for molecular screening and property prediction.

EDUCATION

Ph.D. in Theoretical & Computational Chemistry

Central University of Punjab, Bathinda, India

Thesis submitted; currently under evaluation

Coursework CGPA: 7.80/10.00 (Letter Grade: A - Very Good)

2022-2026

Bachelor of Education (B.Ed.)

The Maharaja Sayajirao University of Baroda, Vadodara, India

CGPA: 7.48/10.00

2019-2020

M.Sc. in Chemistry (Specialization: Organic Chemistry)

The Maharaja Sayajirao University of Baroda, Vadodara, India

CGPA: 7.80/10.00

Dissertation: Studies, Synthesis, and Characterization of Azetidin-2-ones (β -Lactams) Derived from 7-Methyl and 7-Benzoyloxy Derivatives of 2H-1-Benzopyran-2-ones.

Dissertation Supervisor: Dr. A. A. Ajmeri, Associate Professor

2017-2019

B.Sc. (Hons.) in Chemistry

The Maharaja Sayajirao University of Baroda, Vadodara, India

CGPA: 7.97/10.00

2014-2017

TECHNICAL & COMPUTATIONAL SKILLS

Operating Systems:	Linux (Ubuntu), Windows
Molecular Dynamics & Modeling:	GROMACS, CHARMM-GUI, PACKMOL
Electronic Structure & Quantum Chemistry:	Gaussian, ORCA, Multiwfn
Solid-State & Materials Modeling:	VASP, Quantum ESPRESSO
Molecular Docking & Cheminformatics:	AutoDock Vina, Schrödinger Suite, RDKit, Open Babel
Machine Learning & Scientific Computing:	Python, NumPy, SciPy, scikit-learn, TensorFlow, PyTorch, ChemBERTa
Structure Preparation & Visualization:	VESTA, VMD, PyMOL, Avogadro
Programming, HPC & Scripting:	Fortran, SLURM, Bash, awk
Scientific Writing & Plotting:	LaTeX, gnuplot, Grace

FELLOWSHIPS & COMPETITIVE EXAMINATIONS

Council of Scientific and Industrial Research (CSIR-NET), Junior Research Fellowship (JRF): *Chemical Sciences*

Awarded Dec 2019

Graduate Aptitude Test in Engineering (GATE): *Chemistry*

Qualified in 2019 & 2020

Joint Admission Test for M.Sc. (IIT-JAM): *Chemistry*

Qualified in 2017

Joint Entrance Examination - Main (JEE Main): *Physics, Chemistry & Mathematics*

Qualified in 2013 & 2014

PUBLICATIONS

- Singh, P. P.; Dey, C.; Dhal, B.; Muduli, S. B.; Chakraborty, S. In-Silico Scrutinization of (Al, Ga, B, Si, Ge, and P)-Doped C60 in Sensing EGCG: An Application-Based DFT Study. *Comput. Theor. Chem.* 2025, 1244, 115001. [DOI](#)
- Singh, P. P.; Dey, C.; Raphael, A.; Mohammed, F.; Chakraborty, S. Quantum Chemical Investigation of (B, Al, Ga, Ge, Si, N, and P)-Doped C60 in Sensing Ferulic Acid. *Comput. Theor. Chem.* 2025, 1244, 114998. [DOI](#)
- Dey, C.; Singh, P. P.; Chakraborty, S. In-Silico Investigation of the Molecular Disruption of A β 42 Protofibril by Stilbenoids. *J. Biomol. Struct. Dyn.* 2026, 44, 5779-5805. [DOI](#)
- Thomas, J.; Das, A.; Singh, P. P.; Kaur, J.; Kumar, A.; Chakraborty, S.; Jaitak, V. Integrated In Silico Lead Designing Approach for Azepine Derivatives Targeting Human Estrogen Receptor α against Breast Cancer. *Chem. Pap.* 2026, 80, 9535-9563. [DOI](#)
- Dey, C.; Singh, P. P.; Chinyoti, K.; Robins, R.; Chakraborty, S. In-Silico Investigations of Inhibition Capabilities of Selective Estrogen Receptor Modulators against Estrogen Receptor- α (ER- α) in Breast Cancer. *J. Mol. Graph. Model.* 2026, 148, 109496. [DOI](#)
- Chanda, M.; Gangwar, A.; Das, A.; Singh, P. P.; Chakraborty, S.; Jaitak, V. Multi-Step In Silico Study of Potential Estrogen Receptor Alpha Modulators: Integrating Hierarchical Virtual Screening, Pharmacokinetics, MM-GBSA, Molecular Dynamics, and DFT Studies. *J. Mol. Graph. Model.* 2026, 147, 109482. [DOI](#)
- Kumar, A.; Purohit, N.; Singh, P. P.; Jangid, K.; Kumar, V.; Bharat, J. S.; Chakraborty, S.; Kumar, V.; Jaitak, V. Identification of Phytochemicals from Genus Potentilla as Estrogen Receptor- α Inhibitors through Molecular Docking, Molecular Dynamic Simulation and DFT Calculations. *J. Biomol. Struct. Dyn.* 2025, 43, 10167-10183. [DOI](#)
- Das, A.; Sarangi, M.; Jangid, K.; Kumar, V.; Kumar, A.; Singh, P. P.; Kaur, K.; Kumar, V.; Chakraborty, S.; Jaitak, V. Identification of 1,3,4-Oxadiazoles as Tubulin-Targeted Anticancer Agents: A Combined Field-Based 3D-QSAR, Pharmacophore Model-Based Virtual Screening, Molecular Docking, Molecular Dynamics Simulation, and Density Functional Theory Calculation Approach. *J. Biomol. Struct. Dyn.* 2024, 42, 10323-10341. [DOI](#)
- Kumar, V.; Singh, P. P.; Dwivedi, A. R.; Kumar, N.; Kumar, R.; Sahoo, S. C.; Chakraborty, S.; Kumar, V. Caesium Carbonate-Promoted Regioselective O-Functionalization of 4,6-Diphenylpyrimidin-2(1H)-ones under Mild Conditions and Mechanistic Insight. *RSC Adv.* 2023, 13, 16899-16906. [DOI](#)

10. Singh, P. P.; Dey, C.; Kaur, J.; Chakraborty, S. Comparative Analysis of Structural and Dynamical Properties of Lipid Membranes Simulated with the AMBER Lipid21 Force Field Using SPC/E, TIP3P, TIP3P-FB, TIP4P-FB, TIP4P-Ew, TIP4P/2005, TIP4P-D, and OPC Water Models. *bioRxiv* 2026. **(Under Revision)**
11. Singh, P. P.; Kaur, J.; Jakhar, M.; V. C., S.; Dey, C.; Thakran, K.; Chakraborty, S. Hybrid ChemBERTa and DFT Machine Learning Framework for Predicting Enantioselectivity in Organosilanes-Mediated Carbonyl Reduction Reactions. **(Under Revision)**
12. Dey, C.; Singh, P. P.; James, D.; Chakraborty, S. Molecular Mechanism of Phenolic Acid Destabilization Potential on A β Protofibrils. **(Under Revision)**
13. Sahu, P.; Singh, P. P.; Kumar, A.; Chakraborty, S.; Munshi, A. Molecular Dynamics and Simulation Analysis of Pathogenic Variants in Ion Channel-Encoding Genes Associated with Migraine, Epilepsy, and Both. **(Under Revision)**
14. Ghosh, N.; Singh, P. P.; Sharma, A.; Bhattacharya, M.; Chakraborty, S. Stokes-Einstein and Stokes-Einstein-Debye Breakdown in Deeply Supercooled Water: A Comparative Molecular Dynamics Study of Twelve Water Models. **(Manuscript Submitted)**
15. Singh, P. P.; Kumar, A.; Dey, C.; Thakran, K.; Chakraborty, S. DFT-Based Evaluation of Borophene (B36) for Sensing Lenalidomide, Mercaptopurine, and Sulforaphane. **(Manuscript Submitted)**
16. Singh, P. P.; Kumar, B.; Dey, C.; Thakran, K.; Chakraborty, S. Pristine and Al/Ga-Doped Borophene for Antiviral Drug Sensing: A DFT Study of Cidofovir and Tecovirimat. **(Manuscript Submitted)**
17. Singh, P. P.; Laha, P. S.; Dey, C.; Thakran, K.; Chakraborty, S. Rivastigmine Sensing on Pristine, Doped, and Co-Doped Graphdiyne: A DFT Study. **(Manuscript Submitted)**
18. Singh, P. P.; Dey, C.; Chakraborty, S. Benchmarking Water Models for AMBER Lipid14 POPC Bilayers: A Structural and Dynamical Assessment. **(Manuscript Submitted)**
19. Singh, P. P.; Dey, C.; Berlin, N.; Chakraborty, S. Effect of Cholesterol in Regulating SGLT2 Protein's Structural Stability and Lipid Dynamics: Insights from All-Atom Molecular Dynamics Simulation. **(Manuscript Submitted)**
20. Singh, P. P.; Dey, C.; Chakraborty, S. Differential Effects of Structurally Related Hopanoids on DOPC Bilayer Remodeling: An Atomistic Molecular Dynamics Study. **(Manuscript Submitted)**
21. Singh, P. P.; Dey, C.; Chakraborty, S. Structural and Dynamic Behavior of Dipalmitoylphosphatidylcholine Lipid Bilayers in the Presence of Differential Lactulose Compositions: An All Atom Molecular Dynamics Simulation Studies. **(Manuscript Submitted)**
22. Singh, P. P.; Dey, C.; Chakraborty, S. Hydration Structure and Water Dynamics at the Hopanoid-DOPC Membrane Interface. **(Manuscript Submitted)**
23. Singh, P. P.; Dey, C.; Chakraborty, S. Hydration Structure and Water Dynamics at the Lactulose-DPPC Membrane Interface. **(Manuscript Submitted)**
24. Singh, P. P.; Kumari, D.; Dey, C.; Chakraborty, S. Role of Water Model in Molecular Dynamics Simulations of Intrinsically Disordered Protein p53(14-37): Insights from Molecular Dynamics Simulations. **(Manuscript Submitted)**
25. Singh, R.; Singh, P. P.; Prashar, V.; Arora, T.; Chakraborty, S.; Sharma, A.; Parkash, J. Withaferin A Alleviates Reactive Astroglisis by Reducing the Expression of Glial Fibrillary Acidic Protein Induced by Kainic Acid: In Silico to In Vitro Validation. **(Manuscript Submitted)**
26. Dixit, V.; Kalsi, A. S.; Prajapati, V.; Singh, P. P.; Chakraborty, S.; Singh, A. K.; Kumar, P. Deciphering the Antidiabetic Potential of Hibiscus sabdariffa Phytochemicals as SGLT2 Inhibitors: Insights from Molecular Docking, ADME-T, MD Simulation, and DFT Analysis. **(Manuscript Submitted)**
27. Kalra, S.; Singh, P. P.; Bari, V. K.; Chakraborty, S. A Phosphatidylinositol Transfer Protein, PDR16, from Candida auris Clades Involved in Azole Resistance. **(Manuscript Under Preparation)**

† Equal contribution / Joint first authorship.

TEACHING EXPERIENCE

Teaching Assistant - Density Functional Theory (DFT) | Department of Computational Sciences, Central University of Punjab, Bathinda, India

- Assisted with introductory DFT teaching, tutorial discussions, and student support as part of doctoral teaching responsibilities.

CONFERENCES, SYMPOSIA & WORKSHOPS

- **Poster Presentation:** Theoretical Chemistry Symposium (TCS), IIT Bombay, Powai, India, 2-5 December 2025.
- **Poster Presentation:** International Conference on Translational Trends in Natural and Health Sciences (TTNHS 2024), Central University of Punjab, Bathinda, India, December 2024.
- **Poster Presentation:** CECAM Conference MD@60, JNCASR, Bengaluru, India, 26-29 February 2024.
- **Workshop:** CRYSTAL-DFT Modelling in Nanoscience (In-person Workshop with Hands-on Training), INST Mohali, India, 13-16 February 2023.
- **Poster Presentation:** Theoretical Chemistry and Biology (TCB) Symposium, IISER Mohali, India, 15 October 2022.
- **Oral Presentation:** "Mechanistic Investigation of Amine Formation via In Silico Approach," Mini-Symposium on Recent Trends in Biological and Material Sciences, Central University of Punjab, Bathinda, India, 11 May 2022.

AWARDS & RECOGNITIONS

- First Position in Cricket and Football, Chemical Society, Department of Chemistry, The Maharaja Sayajirao University of Baroda, 2017-2018.

ACADEMIC & ADMINISTRATIVE RESPONSIBILITIES

Mess Committee Member | Central University of Punjab, Bathinda, India

- Served for six months, contributing to mess administration, coordination, and drafting of the mess constitution and operational guidelines.

LANGUAGES

Hindi: Native / Mother Tongue | **English:** Professional Working Proficiency (Primary Language of Scientific Communication)

REFERENCES

Dr. Sudip Chakraborty (Ph.D. Supervisor)

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Prof. Anjana Munshi

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Dr. Vikas Jaitak

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